About me

I'm Michael Agazzi a student of the IMT honors program in Learning and Control:

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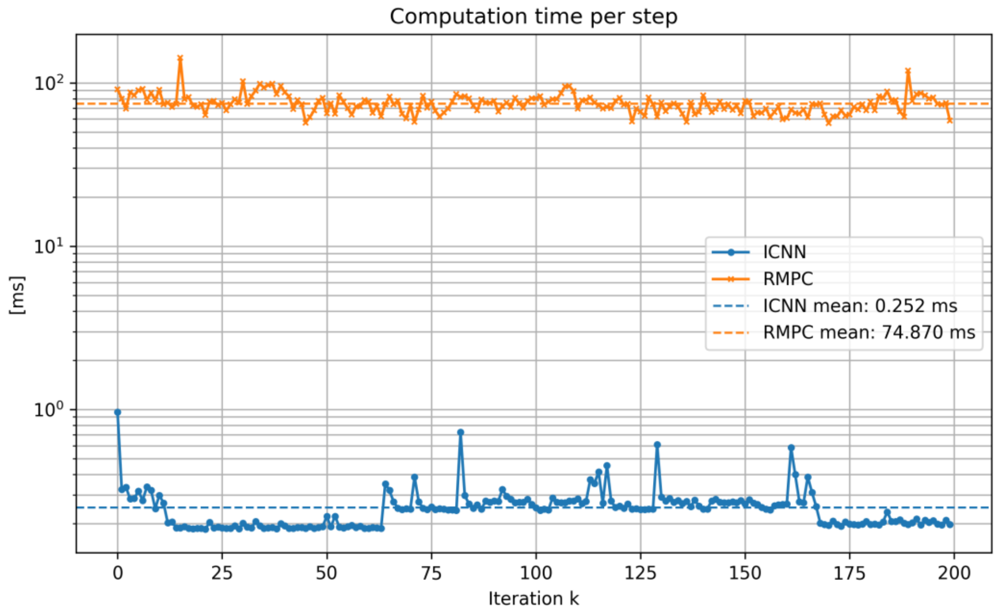
Project: stability guarantee of MPC with neural network approximation of the cost

Motivational example: motor position control

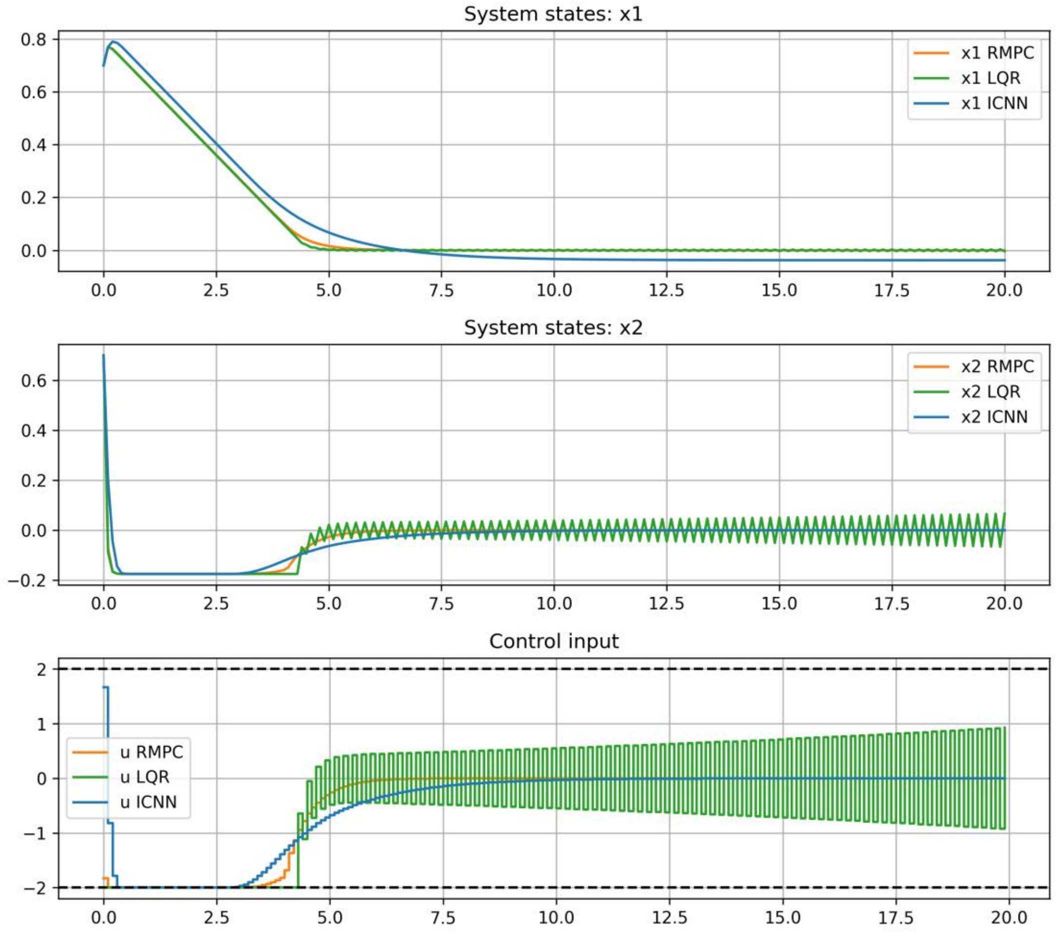
Controller tested:

- LQR: Poor performance
- RMPC: Good performance but slow
- NN: Fast, but with no stability guarantee

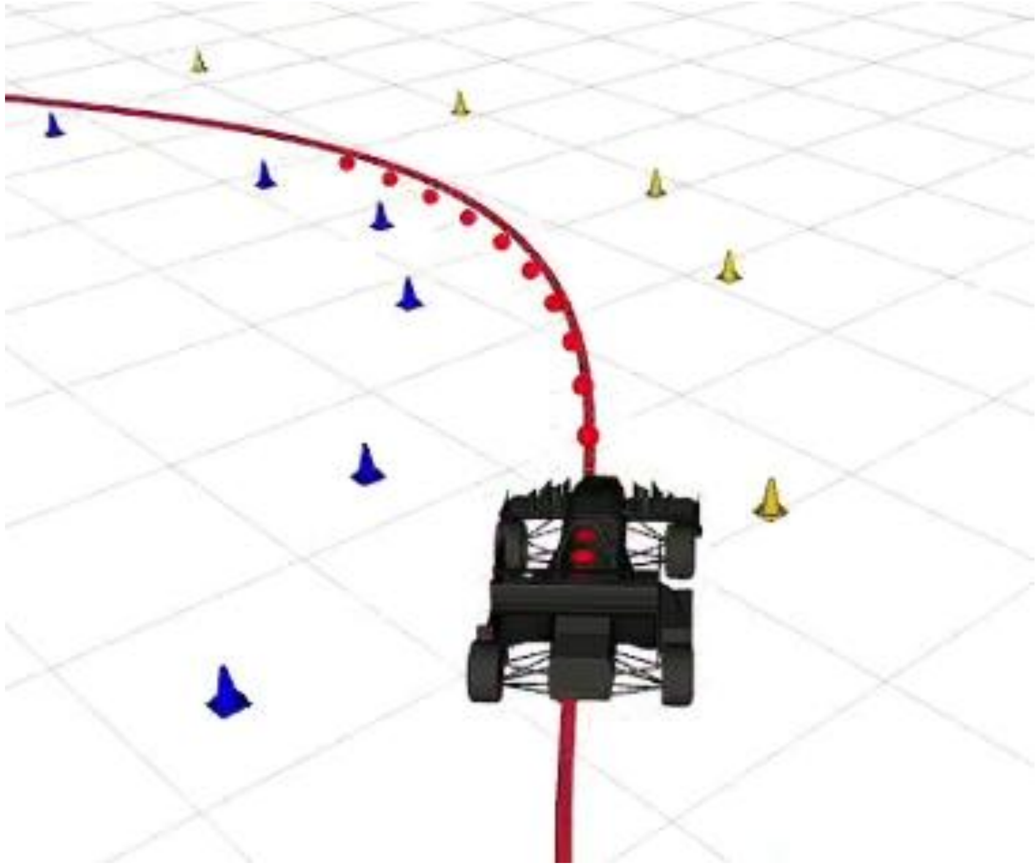


$\alpha(k)$: Parameter proportional to the friction, which is a time-varying parameter

$$x(k+1) = \begin{bmatrix} \theta(k+1) \\ \dot{\theta}(k+1) \end{bmatrix} = \begin{bmatrix} 1 & 0.1 \\ 0 & 1 - 0.1\,\alpha(k) \end{bmatrix} x(k) + \begin{bmatrix} 0 \\ 0.1\,\kappa \end{bmatrix} u(k)$$



Model Predictive Control



$$
\begin{aligned}
\mathcal{J}^N(x_t) = \min_{U_t, X_t} \quad & \sum_{k=0}^{N-1} \ell(x_{k|t}, u_{k|t}) + F(x_{N|t}) \\
\text{s.t.} \quad & x_{0|t} = x_t, \\
& x_{k+1|t} = f(x_{k|t}, u_{k|t}), \qquad k = 0, \ldots, N-1, \\
& u_{k|t} \in \mathcal{U}, \qquad k = 0, \ldots, N-1, \\
& x_{k|t} \in \mathcal{X}, \qquad k = 1, \ldots, N, \\
& x_{N|t} \in \mathcal{X}_f.
\end{aligned}
$$

If $F(x)$ is a control Lyapunov function:

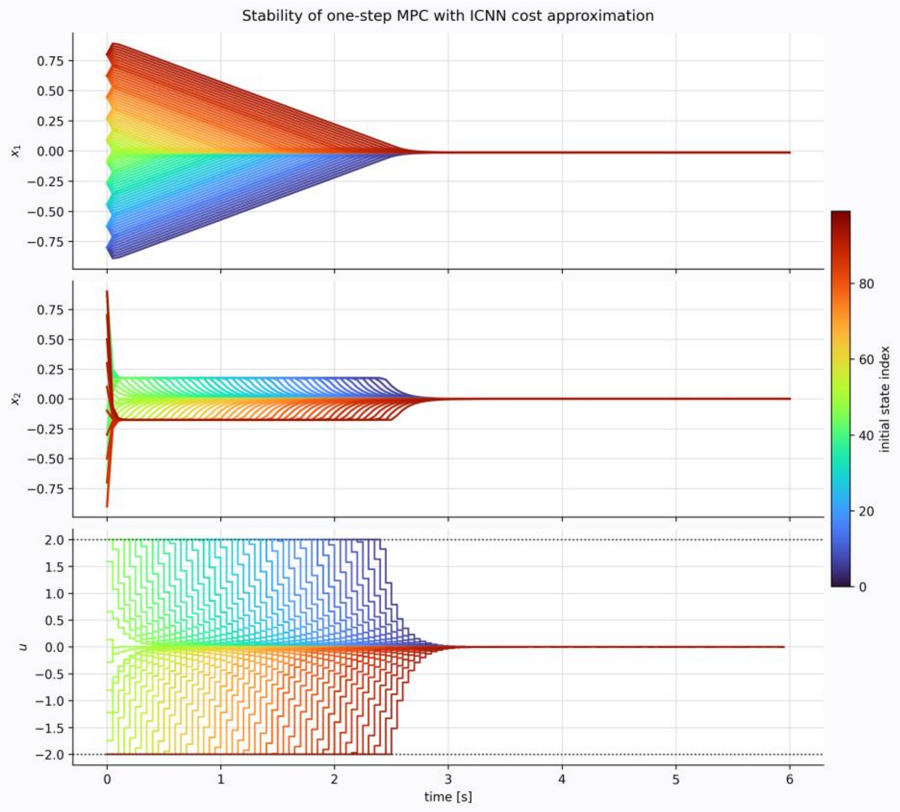
$$\alpha_1(\|x - x_r\|) \leq F(x) \leq \alpha_2(\|x - x_r\|), \qquad \forall x \in \mathcal{X}_f.$$

$$F(f(x, u_T(x))) - F(x) + \ell(x, u_T(x)) \leq 0, \qquad \forall x \in \mathcal{X}_f.$$

the closed-loop system is stable

Model Predictive Control with cost approximation

- It reduces solving time and achieves good performance
- Ongoing: proof the stability outside training data



$$\hat{\mathcal{J}}_\star^1(x_t) = \min_{u_{0|t}, x_{1|t}} \quad \ell(x_{0|t}, u_{0|t}) + \hat{V}(x_{1|t})$$

$$\begin{aligned}
\text{s.t.} \quad & x_{0|t} = x_t, \\
& x_{1|t} = f(x_{0|t}, u_{0|t}), \\
& u_{0|t} \in \mathcal{U}, \\
& x_{1|t} \in \mathcal{X}.
\end{aligned}$$

$$\hat{V}(x_{1|t}) := \min_{U_{1|t}, X_{1|t}} \quad \sum_{k=1}^{N-1} \ell(x_{k|t}, u_{k|t}) + F(x_{N|t})$$

$$\begin{aligned}
\text{s.t.} \quad & x_{k+1|t} = f(x_{k|t}, u_{k|t}), && k = 1, \ldots, N-1, \\
& u_{k|t} \in \mathcal{U}, && k = 1, \ldots, N-1, \\
& x_{k|t} \in \mathcal{X}, && k = 2, \ldots, N.
\end{aligned}$$